

**THE NATIONAL HIGHER SCHOOL OF
ARTIFICIAL INTELLIGENCE**
INTELLIGENT SYSTEMS ENGINEERING DEPARTMENT

SEMESTER 8 – 2025/2026

Reinforcement Learning

Chapter 3: Markov Decision Processes

Prepared by:
Yousra BOUHOUIA

Supervisor:
Hayat Saadi

Exercise 1

Solution

The problem in this exercise is that the reward is too sparse. The robot receives a reward of +1 only when it successfully escapes the maze and 0 at all other time steps.

Using the return definition,

$$G_t = R_{t+1} + R_{t+2} + \dots + R_T,$$

the sequence of rewards in each episode consists of zeros followed by a single +1 at the terminal state. Therefore, if the robot reaches the exit after k remaining steps, the return becomes

$$G_t = \gamma^k,$$

and if the robot does not reach the exit, the return is

$$G_t = 0.$$

At the beginning of learning, the robot rarely escapes from the maze, meaning it mostly receives a total reward of 0. As a result, it does not obtain sufficient feedback to distinguish good actions from bad ones. Consequently, learning is extremely slow and the agent fails to improve its performance.

Exercise 2

Given

Discount factor: $\gamma = 0.5$

Rewards: $R_1 = -1$, $R_2 = 2$, $R_3 = 6$, $R_4 = 3$, $R_5 = 2$

Episode length: $T = 5$

Solution

The objective is to compute the returns $G_0, G_1, \dots, G_5$. The return represents the total discounted future reward from time step t until the end of the episode.

Since the episode terminates at $T = 5$, there are no future rewards after time step 5, therefore:

$$G_5 = 0.$$

We then compute the remaining returns by working backwards using the standard

recursive formula:

$$G_t = R_{t+1} + \gamma G_{t+1}.$$

$$G_5 = 0,$$

$$G_4 = R_5 + \gamma G_5 = 2 + 0.5(0) = 2,$$

$$G_3 = R_4 + \gamma G_4 = 3 + 0.5(2) = 4,$$

$$G_2 = R_3 + \gamma G_3 = 6 + 0.5(4) = 8,$$

$$G_1 = R_2 + \gamma G_2 = 2 + 0.5(8) = 6,$$

$$G_0 = R_1 + \gamma G_1 = -1 + 0.5(6) = 2.$$

Each return is obtained by adding the immediate reward to the discounted value of the next return. Working backwards ensures that all future rewards are correctly incorporated.

Final Answer

$$(G_0, G_1, G_2, G_3, G_4, G_5) = (2, 6, 8, 4, 2, 0).$$

Exercise 3

Given

$$\gamma = 0.9$$

The reward sequence is defined as:

$$R_2 = 2,$$

followed by an infinite sequence of 7s:

$$R_3 = 7, R_4 = 7, R_5 = 7, \dots$$

Solution

We compute the returns using the definition of the discounted return.

Step 1: Computation of G_1

We start by computing G_1 :

$$G_1 = R_2 + \gamma R_3 + \gamma^2 R_4 + \gamma^3 R_5 + \dots$$

Substituting the given values:

$$G_1 = 2 + 0.9(7) + 0.9^2(7) + 0.9^3(7) + \dots$$

We factor out 7 from the infinite part:

$$G_1 = 2 + 7(0.9 + 0.9^2 + 0.9^3 + \dots)$$

The expression inside parentheses is an infinite geometric series with ratio $r = 0.9$. Using the formula for an infinite geometric series,

$$\sum_{k=1}^{\infty} r^k = \frac{r}{1-r},$$

we obtain:

$$\frac{0.9}{1-0.9} = \frac{0.9}{0.1} = 9.$$

Therefore,

$$G_1 = 2 + 7 \times 9 = 2 + 63 = 65.$$

Step 2: Computation of G_0

We now compute G_0 using the recursive formula:

$$G_0 = R_1 + \gamma G_1.$$

Since R_1 is not explicitly specified in the exercise, we express the result in general form:

$$G_0 = R_1 + 0.9 \times 65.$$

If a specific value of R_1 is given, it can be substituted directly into this expression.

Final Results

$$G_1 = 65$$

$$G_0 = R_1 + 58.5$$

Exercise 4

Solution

We want the expectation of the next reward when the current state is $S_t = s$ and actions are selected according to the stochastic policy π .

First, we condition on the action selected by the policy:

$$\mathbb{E}[R_{t+1} \mid S_t = s] = \sum_{a \in \mathcal{A}} \pi(a \mid s) \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a].$$

Then, using the four-argument transition function

$$p(s', r \mid s, a) = \Pr\{S_{t+1} = s', R_{t+1} = r \mid S_t = s, A_t = a\},$$

we compute the inner expectation by averaging over all possible next states s' and rewards r :

$$\mathbb{E}[R_{t+1} \mid S_t = s, A_t = a] = \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} r p(s', r \mid s, a).$$

Substituting back gives:

$$\boxed{\mathbb{E}[R_{t+1} \mid S_t = s] = \sum_{a \in \mathcal{A}} \pi(a \mid s) \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} r p(s', r \mid s, a)}.$$

The reward r must appear as a multiplicative factor (“outside”) because we are computing an expectation (a weighted average): reward value $\times$ probability.

Exercise 5

Solution

1. Equation of v_π in terms of q_π and π :

For any state $s \in \mathcal{S}$, the value of the state is the expected action-value when actions are chosen according to the policy π :

$$\boxed{v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) q_\pi(s, a)}.$$

2. Equation of q_π in terms of v_π and the four-argument p :

Using the four-argument transition probability

$$p(s', r \mid s, a) = \Pr\{S_{t+1} = s', R_{t+1} = r \mid S_t = s, A_t = a\},$$

the action-value function can be written as:

$$q_{\pi}(s, a) = \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) \left[r + \gamma v_{\pi}(s') \right].$$

Exercise 6

Solution

For the gridworld under the (uniform) random policy, the Bellman expectation equation for a non-terminal interior state s is:

$$v_{\pi}(s) = \frac{1}{4} \sum_{a \in \{\uparrow, \downarrow, \leftarrow, \rightarrow\}} \left[r(s, a) + \gamma v_{\pi}(s') \right].$$

For the center state, the value is given as $v_{\pi}(s) = +0.7$ and its four neighboring state values are

$$+2.3, \quad +0.4, \quad -0.4, \quad +0.7.$$

In this gridworld (for interior moves), the immediate reward is $r(s, a) = 0$ and $\gamma = 1$, hence:

$$v_{\pi}(s) = \frac{1}{4} (2.3 + 0.4 - 0.4 + 0.7) = \frac{1}{4} (3.0) = 0.75.$$

Thus,

$$0.75 \approx 0.7,$$

which confirms that the Bellman equation holds *up to the one-decimal rounding* stated in the exercise.

Exercise 7

Solution

The absolute signs of the rewards are not important for comparing states; what matters for preference between states is their *relative* values. In particular, adding a constant c to every reward shifts all state values by the same constant, so the ordering (relative values) of states under any policy is unchanged.

Let the original reward at time $t + 1$ be R_{t+1} and define a new reward

$$R'_{t+1} = R_{t+1} + c.$$

For a fixed policy π , the (discounted) return becomes

$$G'_t = \sum_{k=0}^{\infty} \gamma^k R'_{t+k+1} = \sum_{k=0}^{\infty} \gamma^k (R_{t+k+1} + c) = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} + c \sum_{k=0}^{\infty} \gamma^k.$$

Thus,

$$G'_t = G_t + c \sum_{k=0}^{\infty} \gamma^k.$$

Taking expectation under policy π (starting from $S_t = s$):

$$v'_\pi(s) = \mathbb{E}_\pi[G'_t \mid S_t = s] = \mathbb{E}_\pi[G_t \mid S_t = s] + c \sum_{k=0}^{\infty} \gamma^k = v_\pi(s) + v_c,$$

where the added constant is

$$v_c = c \sum_{k=0}^{\infty} \gamma^k.$$

For $0 \leq \gamma < 1$, the geometric series gives

$$\sum_{k=0}^{\infty} \gamma^k = \frac{1}{1 - \gamma}.$$

Therefore,

$$\boxed{v_c = \frac{c}{1 - \gamma}} \quad \text{and} \quad \boxed{v'_\pi(s) = v_\pi(s) + \frac{c}{1 - \gamma} \quad \forall s.}$$

Since the same constant is added to every state's value, the relative values (and thus the preference ordering of states under any policy) are unchanged.